Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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4 Direct Proof

Use the method of direct proof to prove the following statements.

2. If x is an odd integer, then x^3 is odd.

Proof. Suppose x is an odd integer. Thus $x = 2a + 1$ for some $a \in \mathbb{Z}$. Consequently $x^3 = (2a+1)^3 = 8a^3+12a^2+6a+1 = 2(4a^3+6a^2+3a)+1$. Thus $x^3 = 2b+1$, where b is the integer $4a^3 + 6a^2 + 3a$. Therefore x^3 is odd by definition of an odd number. □

4. Suppose $x, y \in \mathbb{Z}$. If x and y are odd, then xy is odd.

Proof. Suppose x and y are odd. Then $x = 2a + 1$ for some $a \in \mathbb{Z}$, and $y = 2b + 1$ for some $b \in \mathbb{Z}$. Consequently $xy = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$. Thus $xy = 2c + 1$, where c is the integer $2ab + a + b$. Therefore xy is odd by definition of an odd number. □

6. Suppose $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

Proof. Suppose $a \mid b$ and $a \mid c$. Then $b = ak$ for some $k \in \mathbb{Z}$ and $c = ah$ for some $h \in \mathbb{Z}$. Consequently $b + c = ak + ah = a(k + h)$. Thus $b + c = am$, where m is the integer $k + h$. Therefore $a \mid (b + c)$ by definition of divisibility. □

8. Suppose a is an integer. If $5 \mid 2a$, then $5 \mid a$.

Proof. Suppose $5 \mid 2a$. Then $2a = 5k$ for some $k \in \mathbb{Z}$. Observe that $a = 6a - 5a = 3(2a) - 5a$. Substituting $2a = 5k$ gives $a = 3(5k) - 5a = 15k - 5a = 5(3k - a)$. Thus $a = 5n$, where n is the integer $3k - a$. Therefore $5 \mid a$ by definition of divisibility. □

10. Suppose a and b are integers. If $a \mid b$, then $a \mid (3b^3 - b^2 + 5b)$.

Proof. Suppose $a \mid b$. Then $b = ak$ for some $k \in \mathbb{Z}$. Consequently $3b^3 - b^2 + 5b = 3(ak)^3 - (ak)^2 + 5(ak) = a(3k(ak)^2 - ak^2 + 5k)$. Thus $3b^3 - b^2 + 5b = am$, where m is the integer $3k(ak)^2 - ak^2 + 5k$. Therefore $a \mid (3b^3 - b^2 + 5b)$ by definition of divisibility. □

12. If $x \in \mathbb{R}$ and $0 < x < 4$, then $\frac{4}{x(4-x)} \geq 1$.

Proof. Suppose $0 < x < 4$. Then $x > 0$ and $4 - x > 0$. This means that $x(4-x) > 0$, so we can multiply both sides of the inequality by $x(4-x)$, then it is enough to show that $4 \geq x(4-x)$.

Now, $x(4-x) = 4x - x^2$, hence $4 - (4x - x^2) = x^2 - 4x + 4 = (x-2)^2$. Because the square of any real number is positive, $(x-2)^2 \geq 0$, therefore $4 - (4x - x^2) \geq 0$ which implies $4 \geq x(4-x)$. Because $x(4-x) > 0$, dividing both sides by $x(4-x)$ gives $\frac{4}{x(4-x)} \geq 1$. □

14. If $n \in \mathbb{Z}$, then $5n^2 + 3n + 7$ is odd. (Try cases.)

Proof. Suppose $n \in \mathbb{Z}$. Every integer is either even or odd, since dividing an integer by 2 leaves a remainder of either 0 or 1. The following two cases are therefore exhaustive.

Case 1. Suppose n is even. Then $n = 2a$ for some $a \in \mathbb{Z}$. Consequently $5n^2 + 3n + 7 = 5(2a)^2 + 3(2a) + 7 = 20a^2 + 6a + 7 = 2(10a^2 + 3a + 3) + 1$. Thus $5n^2 + 3n + 7 = 2m + 1$, where m is the integer $10a^2 + 3a + 3$. Hence, in this case, $5n^2 + 3n + 7$ is odd by definition of an odd number.

Case 2. Suppose n is odd. Then $n = 2b + 1$ for some $b \in \mathbb{Z}$. Consequently $5n^2 + 3n + 7 = 5(2b+1)^2 + 3(2b+1) + 7 = 5(4b^2 + 4b + 1) + 6b + 10 = 20b^2 + 26b + 15 = 2(10b^2 + 13b + 7) + 1$. Thus $5n^2 + 3n + 7 = 2m + 1$, where m is the integer $10b^2 + 13b + 7$. Hence, in this case, $5n^2 + 3n + 7$ is odd by definition of an odd number.

The two cases are exhaustive and $5n^2 + 3n + 7$ is odd in each, so $5n^2 + 3n + 7$ is odd for every $n \in \mathbb{Z}$. □